

EDUCATION

Ph.D. | Computer Engineering | Polytechnique Montréal & Mila 2021 - Present

- **Research:** Uncertainty in deep learning (variational Bayes, sparse methods), distributional shifts, model robustness, confidence calibration, computer vision.

M.A.Sc. | Engineering Physics | *Graduated with thesis excellence* | Polytechnique Montréal 2016

B.Sc.H. - Co-op | Nanoscience [minors: physics and mathematics] | University of Guelph 2014

PROFESSIONAL EXPERIENCE

Responsible AI Program Manager | AI 2030, USA 2024 - Present

- Leading the design and execution of the Certified Responsible AI Leadership Program (CRAIL), focusing on fairness, accountability, and regulatory compliance.
- Leading discussions with Purdue University, Governance and Responsible AI Lab (GRAIL) Lab and AI 2030 for launching microcredential certification course through Purdue University Online.
- **Awarded:** Program Manager of the Year.

Globalink Mentor | Mitacs Inc., Canada 2023 - 2024

- Mentored and supported Globalink Research Interns (10+), facilitating orientation, providing weekly guidance, and ensuring smooth transitions at host universities.

Metrology Engineer II | Illumina, USA 2018 - 2020

- Developed and integrated new automated image processing algorithms for production QC.
- Improved production process control and tool-to-tool precision from over 50% variability to less than 15%.
- Developed and implemented an image processing pipeline for cross-functional groups, from development to production.
- Led the full cycle of automation and integration of a new metrology technique to production within the framework of Software Lifecycle Management, from defining user requirements to reviewing and validating test cases and production release.
- **Awarded:** FoFee Award for Technical Excellence in driving software development and automation for NPI projects.
- **Awarded:** for exceptional software development leadership and significant image analysis QC improvements.

Metrology Scientist | Teledyne MEMS, Canada 2016 - 2018

- Led the materials analysis, optical metrology, and characterization for ultra-thin films (PVD/CVD) for pellicle development for EUV Lithography with ASML for enabling next generation 3 nm transistors by TSMC.
- Characterized and optimized thin film surfaces and interfaces via XPS, AFM, SEM, and XRR techniques.
- Developed ellipsometry models for PVD and CVD films for in-line process monitoring with GageR&R below 10% for single and multi-layer ultra-thin films (<10 nm).
- Designed and executed DOEs for root cause analysis, identified a critical workflow problem, and implemented a major process change. This improved product performance and eliminated specific scrap recurrence, achieving zero defect recurrence related to the specific failure mode.
- Designed and executed DOEs for the development of process-product correlations.
- **Distinction:** Nominated for Employee of the Year Award in Technology Innovation.

RESEARCH EXPERIENCE

Doctorate Research Assistant | Polytechnique Montréal, Canada 2021 - Present

- Research in deep learning model uncertainty and generalizability for computer vision applications.
- Sparsity in variational Bayes and deterministic neural networks for parameter-efficient solutions.
- Evaluation of risks of covariate shifts in real-world data and impact on model generalizability.
- Multiple first-author publications, including two in top-tier conferences in medical image analysis and machine learning.

- **Awarded** NSERC Postgraduate Scholarship-Doctoral (PGS-D), converted to CGS-D. National Competition.
- **Awarded** FRQNT PhD Scholarship. Ranked no.1 in Techniques, Measurements, & Systems committee. Provincial Competition.
- **Awarded** Travel & Registration award to 38th NeurIPS Conference by Women in Machine Learning (WiML) Workshop.

Research Assistant | San Diego State University, USA 2019

Masters Research Assistant | Polytechnique Montréal, Canada 2015 - 2016

- Characterization of gas-solid interaction in nano-getter material for integrated non-evaporable getter in wafer-level packaged microbolometers developed by Teledyne MEMS.
- First-authored journal article, a poster presentation, and an invited talk at an international symposium.
- **Awarded** Mitacs Graduate Student Scholarship. National competition.
- **Awarded** Best Master's Thesis in Engineering Physics.
- **Awarded** Best poster award at 10th NAMIS International Research Network, University of Tokyo, Japan.
- **Distinction** Nominated for Best Thesis Award at institutional level.

Postgraduate Research Assistant | University of Guelph, Canada 2014

Undergraduate Research Assistant | FPIInnovations, Canada 2013

Undergraduate Research Assistant | University of Alberta, Canada 2013

Undergraduate Research Assistant | University of Notre Dame, USA 2012

Undergraduate Research Assistant | TRIUMF, Canada 2011

PUBLICATIONS

Conference Proceedings

- **Abboud, Z.**, Lombaert, H., and Kadoury, S. Variational Visible Layers: A Practical Framework for Uncertainty Estimation. **(2025)**. MICCAI. Oral. (29% acceptance rate, 7% Oral.) [link](#)
- **Abboud, Z.**, Lombaert, H., and Kadoury, S. Sparse Bayesian Networks: Efficient Uncertainty Quantification in Medical Image Analysis. **(2024)**. MICCAI. (early accept 11% of submissions, 30% overall acceptance rate). [link](#)
- **Abboud, Z.** and Kadoury, S. Impact of train- and test-time Hounsfield unit window variation on CT segmentation of liver lesions. **(2023)**. SPIE Medical Imaging. [link](#)

Journal Articles

- Assali, S., Koelling, S., **Abboud, Z.**, ..., Moutanabbir, O. **(2022)**. Journal of Applied Physics. [link](#)
- Yang, F., ..., **Abboud, Z.**, ... and Gu, J. **(2020)** Chemistry-Sustainability-Energy-Materials (ChemSusChem). [link](#)
- **Abboud, Z.** and Moutanabbir, O. **(2017)** J. Physical Chemistry C. [link](#)
- **Abboud, Z.**, ... and Mohanty, A. K. **(2016)**. Materials Letters. [link](#)
- George, M., Mussone, P. G., **Abboud, Z.**, and Bressler, D. C. **(2014)**. Applied Surface Science. [link](#)
- Teigelhoefer, A., Lassen, J., **Abboud, Z.**, ... and Raeder, S. **(2013)**. [link](#)

Invited Talks

- **Abboud, Z.**, ... Moutanabbir, O. **(2018)**. The ElectroChemical Society (ECS): Americas International Meeting on Electrochemistry and Solid State Science (AiMES). [link](#)
- **Abboud, Z.**. Materials Science in Modern MEMS: Challenges and Opportunities. **(2017)**. RQEMP.

TECHNICAL REVIEWER

CVPR Main Conference	2025	NeurIPS WiML workshop	2024
IEEE Transactions on Medical Imaging	2024, 2025	ICML WANT workshop	2024
MICCAI Main conference	2024, 2025	NeurIPS WANT workshop	2023

OPEN SOURCE CONTRIBUTION

Pytorch Bug Sparse tensor multiplication - bug report and resolution	2023-Present
Pytorch Docathon Ranked second (H1) & third place (H2)	2023-2024
Pytorch Documentation SparseAdam optimizer documentation	2023

TECHNICAL & PROFESSIONAL SKILLS

Programming Python	Metrology Gage R&R, Optical Metrology
Tools PyTorch, Git, OpenCV, JMP	Spectroscopy, Microscopy, & Spectrometry XPS, AES, XRR, XRD, AFM, SEM, TEM, TOF-SIMS
Data Management SQL, Postgres, AWS	Process Engineering Performance Correlation, Failure Analysis, Statistical Process Control, MES
Software Lifecycle Management User Requirements, Software Quality, Testing & Validation	

VOLUNTEER EXPERIENCE

Communication & Technology Chair WiML Workshop, NeurIPS	2025
Super Volunteer WiML Workshop, NeurIPS	2024-2025
HBHL Symposium Volunteer McGill University, Montréal, Canada	2023
Poly-Fi Mentor Volunteer Polytechnique Montréal, Montréal, Canada	2022 - 2023
When Tomorrow Comes Benefit Show Volunteer Montréal, Canada	2016
Graduate Student Week Volunteer Polytechnique Montréal, Montréal, Canada	2016
New Graduate Student Event Volunteer Polytechnique Montréal, Montréal, Canada	2016
McGill Innovation Week Volunteer McGill University, Montréal, Canada	2015
Undergraduate Lab Tour Volunteer Polytechnique Montréal, Montréal, Canada	2015
ICSI Conference Volunteer Polytechnique Montréal, Montréal, Canada	2015
Science and Engineering Sunday Volunteer University of Guelph, Guelph, Canada	2014
BioNIB Meeting Volunteer University of Guelph, Guelph, Canada	2014
CPES Undergraduate Poster Judge University of Guelph, Guelph, Canada	2014
Alumni Weekend BDDC Tour Volunteer University of Guelph, Guelph, Canada	2014
ISBBB Conference Volunteer University of Guelph, Guelph, Canada	2014
Academic Open House University of Guelph, Guelph, Canada	2014